Prediction of Dementia Status Using Longitudinal Studies

CONOR MURPHY

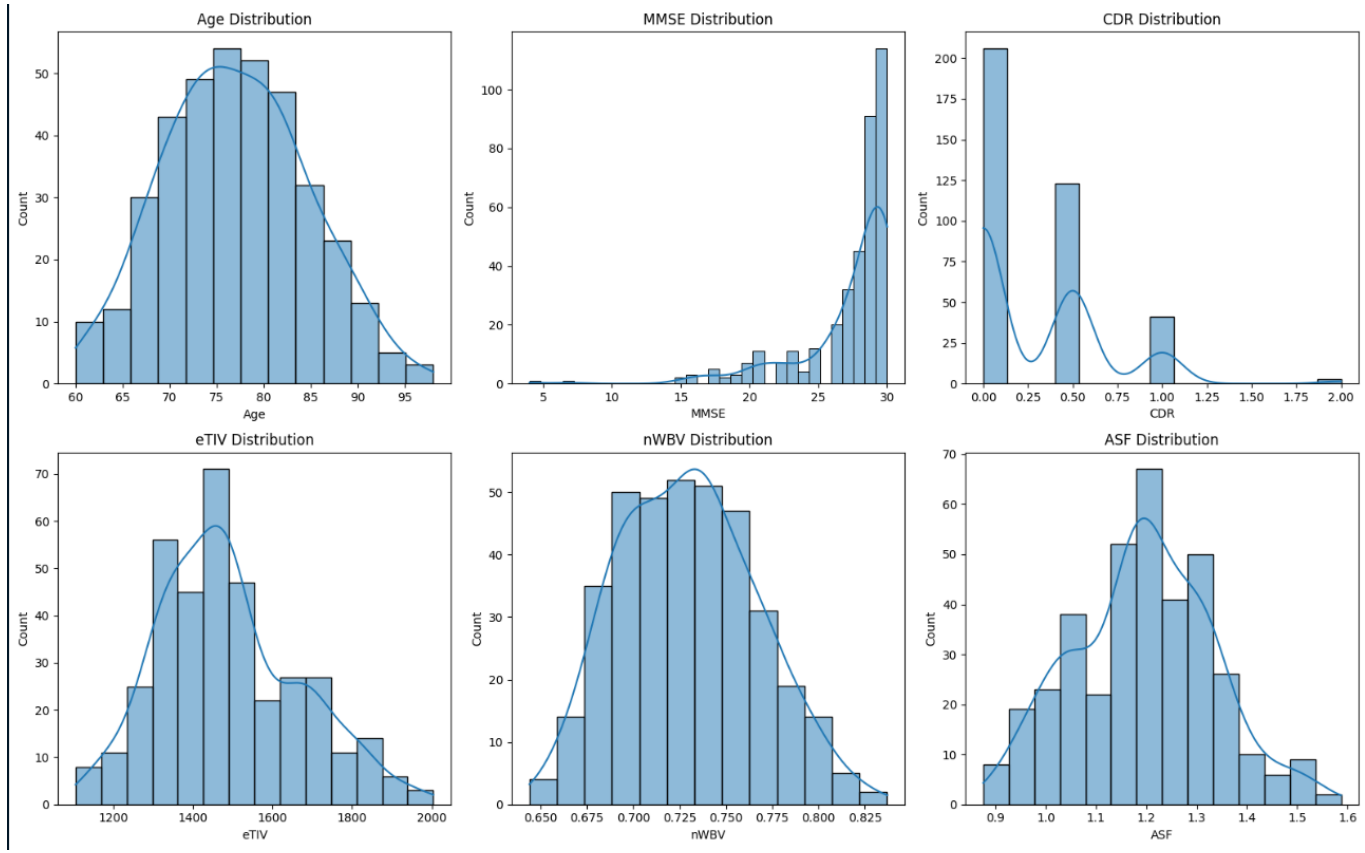
Data Description and Classes to Predict

- The dataset contains longitudinal data from 150 patients, all age 60 or over, who have undergone multiple MRI scans with a time period of more than a year between visits.

- Our goal is to predict the class label associated with dementia progression states over time:
- 1. Nondemented – Nondemented
- 2. Demented – Demented
- 3. Nondemented – Demented (Converted)

- Key Data Characteristics:
- Each patient has multiple visits with multiple measurements taken each time
- Key features include: cognitive scores (MMSE), MRI-Metrics (eg. nWBV), demographic data (e.g Age), and clinical ratings (e.g CDR)
- Challenge: Using the longitudinal nature of the data to predict transitions between dementia states. Critical when predicting progression of dementia.

Data Feature Distribution



Data Pre-Processing Approach

- Pre processing was handled in a variety of ways.

- Ensuring usability: forward fill – ‘ffill’ method used to ensure continuity of the data per individual e.g for the socioeconomic status column that had missing data. Hand column dropped due to homogeneity.

- Encoding categorical variables such as Group and M/F to transform into numerical formats.

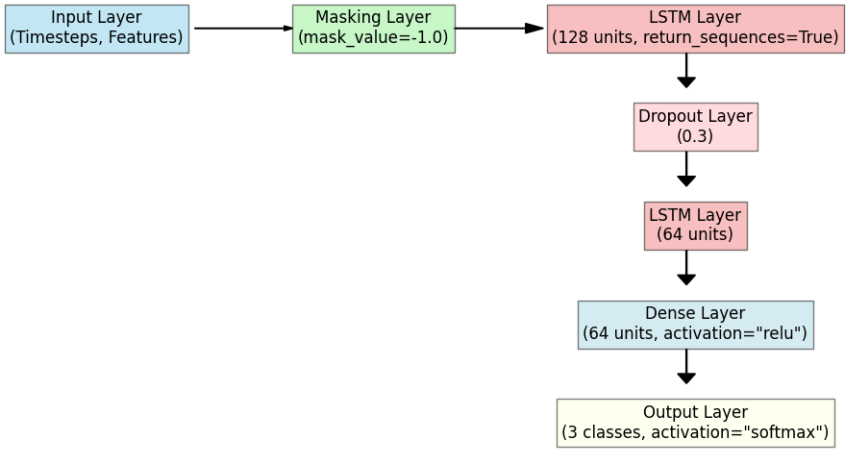
- Standardisation of continuous variables was achieved using StandardScaler: Age, MR Delay, nWBV to ensure no single feature dominates. Having a mean of 0 and SD 1 is important for models such as Logistic Regression and Support Vector Machines used in the ensemble method.

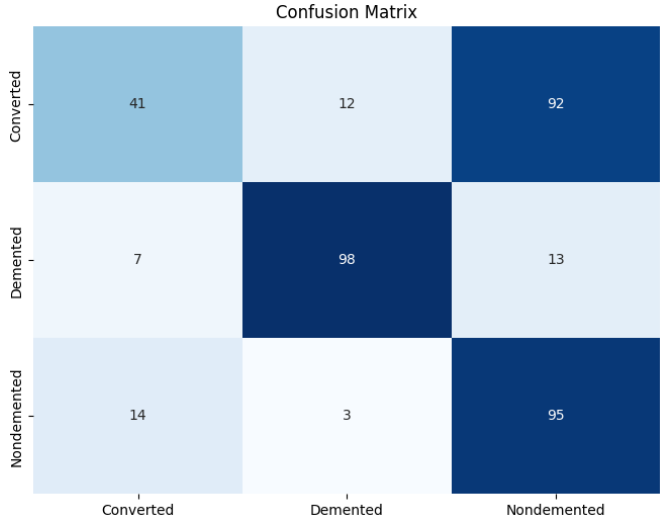
Data Transformation and Balancing

- As the data is already in the required “Long” format with each row corresponding to a specific visit, this transformation was not necessary.
- However transformation was required to change the structure of the data to be suitable for models such as LSTM, which needs a sequence of feature vectors over time
- The data was grouped by Subject ID and sorted by Visit. Then sequences were created using the selected features Age, MMSE, CDR, MR Delay, and nWBV.
- Padding was also used to ensure sequences all have the same length, a requirement of the LSTM model.
- SMOTE was used to balance the class distribution by creating synthetic samples due to the minority class ‘Converted’ being underrepresented

Analysis Methodology – LSTM

- LTSM was chosen for its ability to capture long term dependencies in sequential data, making it ideal for our longitudinal dataset.
- Initial masking layer to handle varied sequence lengths, followed by two LSTM layers 128-64 units. Dropout rate set to 30% to prevent overfitting.
- Dense Layers - One (64 units with ReLu) which adds nonlinearity to achieve lower training loss and a softmax output layer to handle multiple classes.
- TimeSeriesSplit validation was employed to ensure predictions were evaluated realistically by training on past data and testing on future data, maintaining the temporal sequence.
- Results: Low prediction of converted cases but precision remained high. Model may be too permissive/biased for nondemented classification at 0.48 precision. Micro-average ROC AUC was calculated at 0.9. Suggests on avg performing well at classifying, but a high ROC AUC compared to the results may also support overfitting or bias in the model.





Class	Precision	Recall	F1-Score	Support
Converted	0.65	0.28	0.39	145
Demented	0.88	0.83	0.85	118
Nondemented	0.48	0.85	0.61	112

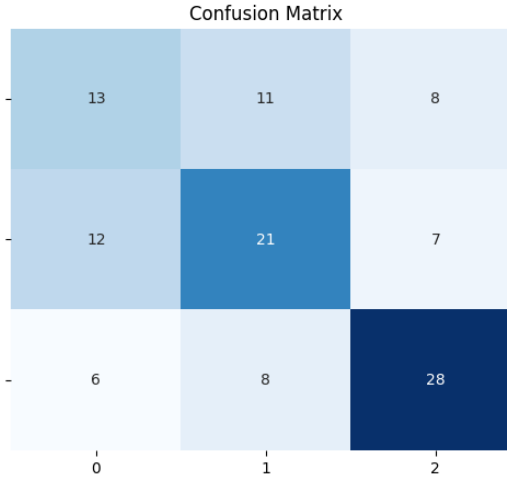
Analysis Methodology – Mixed Effects Model

- Mixed Effects Models are well suited for longitudinal data as they capture both fixed (time, age) effects and random (individual variability) effects.
- Dependent variable was the Clinical Dementia Rating.
- Fixed Effects of Visit, Age, Sex, MR Delay, nWBV
- Random Effect of subject level intercepts and slopes, which account for individual variability in dementia progression.
- Results: nWBV negatively correlated with CDR, whilst Age positively correlated with CDR, indicating these as key factors in dementia progression

Predictor	Coefficient	Std. Error	p-value	95% CI
Intercept	0.207	0.073	0.005	[0.064, 0.350]
Sex (M/F)	0.076	0.052	0.145	[-0.026, 0.178]
Visit	0.044	0.037	0.236	[-0.029, 0.116]
Age	0.092	0.029	0.002	[0.035, 0.149]
MR Delay	0.009	0.033	0.782	[-0.055, 0.073]
nWBV	-0.141	0.027	0.0	[-0.195, -0.088]

Analysis Methodology – Ensemble Model

- As part of exploring multiple avenues, an Ensemble Model using a Voting Classifier, which combined Logistic Regression, SVM, and Random Forest was also used.
- This was explored less than the other two methods as it does not inherently capture the longitudinal form of the data as well as MRMs or LSTMs. However, it was used to guide the expected results in terms of classifying the Converted class, and was inspired by Battineni et al., (2020) and their ensemble approach to predict Alzheimers.
- The individual models in the ensemble treat data as independent observations and do not naturally account for the temporal nature.
- Results: Micro Avg ROC AUC – 0.79



Class	Precision	Recall	F1-Score
Converted	0.42	0.41	0.41
Demented	0.53	0.53	0.53
Non-Demented	0.65	0.67	0.66

Ref: Battineni, G. *et al.* (2020) 'A comprehensive machine-learning model applied to magnetic resonance imaging (MRI) to predict alzheimer's disease (AD) in older subjects', *Journal of Clinical Medicine*, 9(7), p. 2146. doi:10.3390/jcm9072146.

Model Performance Comparison

- LSTM – High ROC AUC likely indicative of high discriminative ability. Captures temporal dependencies well due to sequential nature, best at precisely distinguishing converted cases. However high complexity, possibility of overfitting and low precision for nondemented cases hold it back.
- Mixed Effects Model – Offers insight into fixed and random effects. Provides clear interpretability and shows statistically significant predictors like nWBV. However, has limited predictive power compared to the other two methods, primarily used for understanding relationships.
- Ensemble – High ROC AUC, although lower than LSTM by 0.11, showing good overall performance. Balances predictions by using multiple algorithms so strong against overfitting. However has lower recall and precision than LSTM for most classes, and is overall less effective in capturing temporal dynamics.

Discussion and Conclusion

- Demonstrates the efficacy of using LSTM for longitudinal data. Capturing temporal patterns has proven beneficial for predicting progression to the converted or newly captured dementia class.
- LSTM networks, when combined with data preprocessing and SMOTE, can be a promising approach for predicting dementia progression.
- Mixed effects model identified key predictors such as Age and nWBV which are useful for understanding dementia progression, and are key factors to look out for in terms of patient clinical care. Patients displaying such correlated risk factors could be called in for more frequent monitoring/MRI's and reduce their MR Delay.
- Challenges included handling class imbalance, with many models initially not classifying converted at all, sequence length variability, and model complexity. As seen in work by Battineni (2019), classifying converted type is difficult with many non demented subjects being incorrectly forecasted and 0 actually being classified as True Converted.
- In future, hybrid models combining LSTM with methods employed in the ensemble could prove more accurate.

Ref: Battineni, G., Chintalapudi, N. and Amenta, F. (2019) 'Machine learning in medicine: Performance calculation of dementia prediction by support vector machines (SVM)', *Informatics in Medicine Unlocked*, 16, p. 100200. doi:10.1016/j.imu.2019.100200.